Labwork 3
Logistic Regression

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1 Logistic Regression

This section is about the implementation of Logistic Regression from scratch.

1.1 Gradient Descent

Gradient Descent Algorithm to find the minimize value of the function $f(x)$:

$$x_{i+1} = x_i - lr \times f'(x)$$

1.2 Sigmoid Function

The sigmoid function is used to transform the output of the last layer into a value between 0 and 1, which can be interpreted as a probability:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

1.3 Loss Function

The loss function is calculated by using binary cross-entropy:

$$\text{Loss} = -\frac{1}{n} \sum_{i=1}^{n} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

1.4 Implementation

Firstly, I initialize the parameters $w = [\mathbf{1,\ 1,\ 1}]$. The model was running on 100 epochs with learning rate 0.01. I also applied the epsilon to stop training when the difference between two consecutive loss values is less than 0.001.

Epoch	Loss
1	3.1958
2	1.2224
3	0.6405
4	0.5984
	⋮
22	0.5724
23	0.5714

Table 1: Loss during training

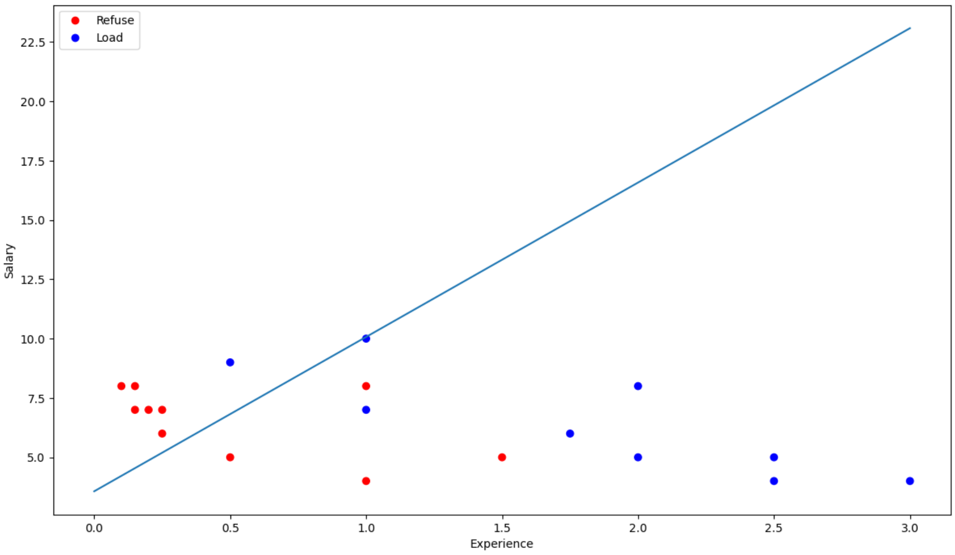

Figure 1: Prediction line